

forage – Product Requirements Document (PRD)

Product Summary

forage is a spatial inspiration capture system that allows designers to gather inspiration directly from the physical world using gesture interactions. Captured inspiration becomes structured digital design assets that are automatically organized by AI and curated by users inside a web-based creative archive. The system bridges real-world observation with digital creative workflows, enabling designers to capture, rediscover, and reuse inspiration efficiently.

Problem Statement

Designers encounter inspiration constantly in the real world, but current capture methods are fragmented. Photos accumulate in camera rolls, screenshots pile up in folders, and references become difficult to retrieve later during design work. Designers need a system that captures inspiration instantly, organizes it automatically, and allows fast retrieval of usable creative assets.

Product Vocabulary

Concept	Product Language
Item	Capture
Feed	Field
AI Groups	Findings
File-Type Browsing	Types
Project Boards	Boards

Core Product Principles

Principle	Meaning
Capture at the Speed of Perception	Inspiration should be captured instantly.
Gesture-Based Interaction	Natural gestures control capture.
Minimal Interaction Complexity	A small set of intuitive gestures.
AI-Assisted Organization	Captures automatically grouped into Findings and Types.

Hardware Component – Capture Ring Module

Property	Description
Attachment	Snap-on module that attaches to an existing ring
Placement	Back of the ring band

Activation	Double tap ring to activate or deactivate capture mode
Capture Method	Gesture interactions performed after outlining capture region

Simplified Gesture System

Step	Gesture	Interaction	Intent	Output
1	Outline in Air	Draw loose outline around object	Define capture region	Selection canvas
2	Tap	Tap after outlining	Capture image reference	PNG / JPG
2	Two-Finger Tap	Tap with two fingers	Capture color	HEX color code
2	Two-Finger Drag	Drag across text	Capture typography	OCR text + font match
2	Hold + Drag	Hold and drag slightly	Capture motion	MP4 + GIF preview + timing curve

Motion Capture System

Motion capture allows designers to record real-world movement and convert it into usable animation references for motion graphics workflows.

Output	Purpose
MP4 Motion Clip	Primary motion reference for animation work
GIF Preview	Looping preview displayed inside the archive interface
Animation Timing Curve (JSON)	Extracted easing behavior used for motion timing references

Web Application Architecture

Area	Purpose
Field	Archive of gathered inspiration sorted by recency
Findings	AI-generated clusters revealing patterns across captures
Types	File-type browsing system allowing retrieval by asset format
Boards	Project-specific curated inspiration collections
Search	Global search across captures, findings, and boards

AI Categorization System

File Type	AI Category Examples	Purpose
Typography	Serif, Sans-Serif, Monospaced, Display	Helps designers locate type styles
Images	People, Food, Architecture, Street, Nature	Groups visual scenes automatically
Motion	Ambient motion, looping motion, mechanical motion	Helps categorize motion references
Colors	Single colors, palettes, gradients	Helps discover color references
Textures	Metal, paper, fabric, glass, concrete	Groups material references

Color Capture System

The system supports two color capture types: single color captures and palette captures extracted from images. Single colors appear as individual swatches with HEX values, while palette captures display 4–6 extracted colors representing the scene.

Capture Detail Page

Clicking any capture opens a dedicated capture detail page displaying a large preview of the capture along with metadata, downloadable outputs, and related captures. Designers can download assets such as MP4 motion clips, copy color values, inspect timing curves, or add captures directly to Boards.

Core User Flow

- User double taps ring to activate capture mode
- User outlines region in air
- User performs capture gesture
- System extracts capture
- Capture saved to Field
- AI groups captures into Findings and Types
- User organizes inspiration into Boards